FPGA Tactical Controller for StarCraft II Micro Scenarios

Target Tier: Tier A (100%) **Course:** EECS 3216 (Fall 2026) **Topic:** Datapath / Control / VGA

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1. Overview and Difficulty Justification

In the StarCraft II (SC2) mini-game *DefeatRoaches*, friendly Marines fight waves of enemy Roaches. [built] A PC sends each game snapshot to the DE10-Lite over UART; the FPGA chooses one attack target per Marine and replies, the PC issues those orders, and VGA draws the units and target lines. [proposed] A Lanchester strength check and a tactical FSM will also decide *when to disengage*: ATTACK, RETREAT or HOLD. Tier A: the UART link, ranking pipeline, FSMs and VGA run concurrently in group-written HDL.



Adapted from Lee and Seshia, Fig. 1.1 [1]: SC2 replaces the physical plant, and its observation and order APIs act as sensors and actuators. The PC only relays; every targeting and tactical decision is made on the FPGA.

2. Target Hardware and Peripherals (DE10-Lite Only)

- **Onboard:** MAX 10 FPGA (`10M50DAF484C7G`) at 50 MHz; 2×20 GPIO header (GPIO[0]/[1] = UART RX/TX); VGA port (4-bit RGB); SW4:0; KEY0; HEX5:0; LEDR9:0.
- **External:** 3.3 V USB-to-UART adapter with common ground; 640×480 VGA monitor; PC running SC2 with a Python adapter (`burnysc2`, `pyserial`).

3. Combinational and Sequential Logic Architecture

- **Combinational:** [built] subtractors and multipliers compute the exact squared Marine–Roach distance (no square root); comparators test attack range and compare ranking keys; an 8-bit adder checks each frame's checksum. [proposed] A 17-entry square table and integer comparators evaluate the strength check (Section 4).
- **Sequential:** [built] UART RX/TX FSMs and packet framers; snapshot registers; a three-stage distance/ranking pipeline that accepts one Marine–Roach pair per clock, so one decision takes exactly $n_F n_E + 2$ cycles (at most 258 cycles, 5.16 µs); VGA timing counters. [proposed] A tactical FSM (ATTACK, RETREAT, HOLD) with a retreat latch, run after target ranking.

4. Proposed Extension: Lanchester Strength Check

In Lanchester's square law, a side's strength in a ranged fight grows with the square of its unit count; Stanescu et al. show such attrition laws predict StarCraft combat [3]. We use only the square-law strength, not their fitted model: $P_F = k_F N_F^2$ and $P_E = k_E N_E^2$, where $N \in \{0, \ldots, 16\}$ counts living units and k is a fixed 16-bit coefficient per side for this matchup (damage rate against opponent health). N^2 comes from the table, so products fit in 24 bits with no square root or division. **Rule:** latch RETREAT if $4P_F < 3P_E$ (friendly strength below 75% of enemy) or friendly HP falls to 25% of its starting total; otherwise ATTACK. HOLD when either side is empty or the retreat reaches its safe point; the latch clears only on reset or a new encounter. A new protocol version adds friendly HP to the request and state, reason and both strengths to the reply. Cutoffs are our engineering parameters, not from [3]; the model ignores range, wounds and waves, so retreat is judged by Marines preserved.

5. Concrete Specifications and Testable Features

- **SC2 targeting and tactical decisions:** [built] SW0=0: each Marine attacks the nearest Roach. SW0=1, focus fire [2]: the weakest Roach in range, otherwise the nearest. [proposed] ATTACK/RETREAT/HOLD orders from Section 4. *Pass:* board replies match an independent Python reference on every snapshot; retreat preserves more Marines than a no-retreat baseline.
- **VGA visualization of controller state:** [built] 640×480 at 60 Hz showing units, Marine-to-target lines and policy; HEX5:4 snapshot ID, HEX3 policy, HEX2 status, HEX1:0 the target of the Marine chosen by SW4:1; LEDR shows policy, receive, error and distinct-target count. [proposed] Strength bars, tactical state and reason. *Pass:* for a frozen snapshot, the display matches the reply frame.
- **UART communication:** [built] 115200 baud 8-N-1, one request and one reply, up to 16 units per side (151-byte request, 13.1 ms). Checksum, count, range and 10 ms timeout checks; a bad frame gets an error status and never changes the display. *Pass:* a corrupted frame is rejected and the next valid frame is processed without reset; every reply reports datapath cycles.
- **Originality:** all HDL (UART, framers, pipeline, FSMs, VGA) is group-written; the PC adapter uses `burnysc2`/`pyserial` and Blizzard's map.

6. Verification Plan and Team Task Allocation

Marlin Lachance (lead): existing pipeline, architecture, tactical FSM, SC2 host integration. **Vaughn Chan (hardware):** board setup, UART/VGA verification, strength and state displays. **Hafsa Manazir (documentation):** report, references, results. **Vladislav Prasko (demonstration):** SC2 scenarios, integration testing, evidence, presentation. We also report Quartus timing/resources, datapath cycles and UART latency, and claim no CPU speedup (the 13.1 ms link dominates).

[1] E. A. Lee and S. A. Seshia, *Introduction to Embedded Systems: A Cyber-Physical Systems Approach*, 2nd ed., MIT Press, 2017, Fig. 1.1.
[2] D. Churchill and M. Buro, "Portfolio Greedy Search and Simulation for Large-Scale Combat in StarCraft," *IEEE CIG*, 2013, Sec. III.
[3] M. Stanescu, N. Barriga and M. Buro, "Using Lanchester Attrition Laws for Combat Prediction in StarCraft," *AIIDE*, 2015.